s7_directional

equity strategy | family: equity-directional | status: reference | derived from `s6_neutral_levered`

What this strategy does

Strategy 6 signal, cells neutralized across sides together, raw returns

Why it should work. Let net exposure float with the signal; empirically this smuggles in uncontrolled market beta.

Design notes (from the strategy's source file).

Strategy 7 -- Unconstrained net exposure (cautionary result).

Strategy 5/6's signal, but the long and short legs are no longer forced to equal dollar totals, and returns are RAW so real market beta is carried. Net dollar exposure is only ~10% of gross (as in every other book) yet the book is 0.52-correlated with the market, because raw returns carry each stock's own beta and equal dollars long and short are not equal BETA: high-SUE and low-SUE firms differ in beta. Kept as a documented counter-example to the explicit beta overlay.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	priority
beta_overlay	0.0

Performance

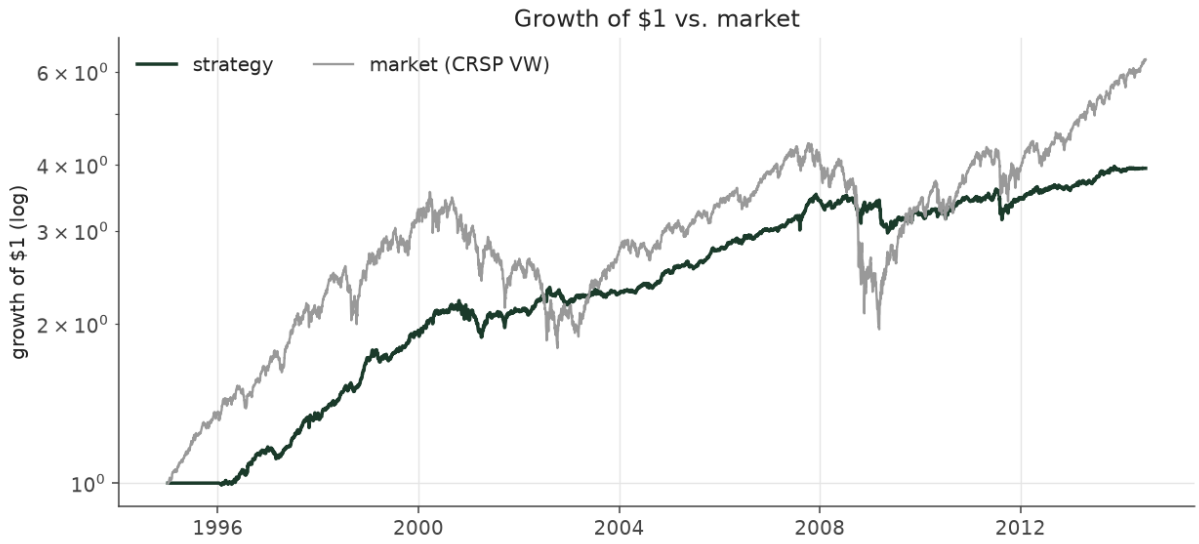
metric	s7 directional	market (CRSP VW)
Annualized return	7.30%	9.94%
Annualized volatility	6.56%	19.35%
Sharpe (NAV-pct)	1.11	0.59
Sortino	1.61	
Max drawdown	-15.5%	-55.5%
Calmar	0.47	
Final NAV ($10M start)	$39,420,102	
Worst day	-3.54%	
% positive days	51.7%	
Sharpe 1996-2001	1.79	
Sharpe 2002-2007	1.48	
Sharpe 2008-2013	0.32	

Correlation of daily returns with the market: **0.52**.

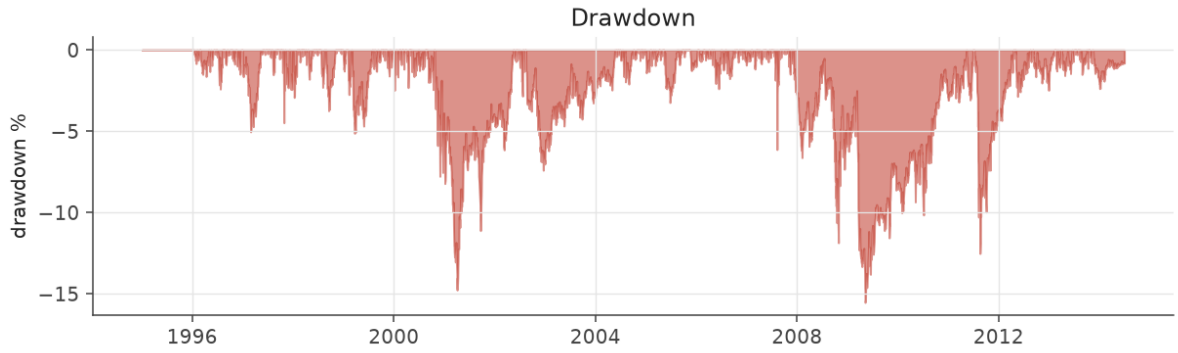
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.11

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.073

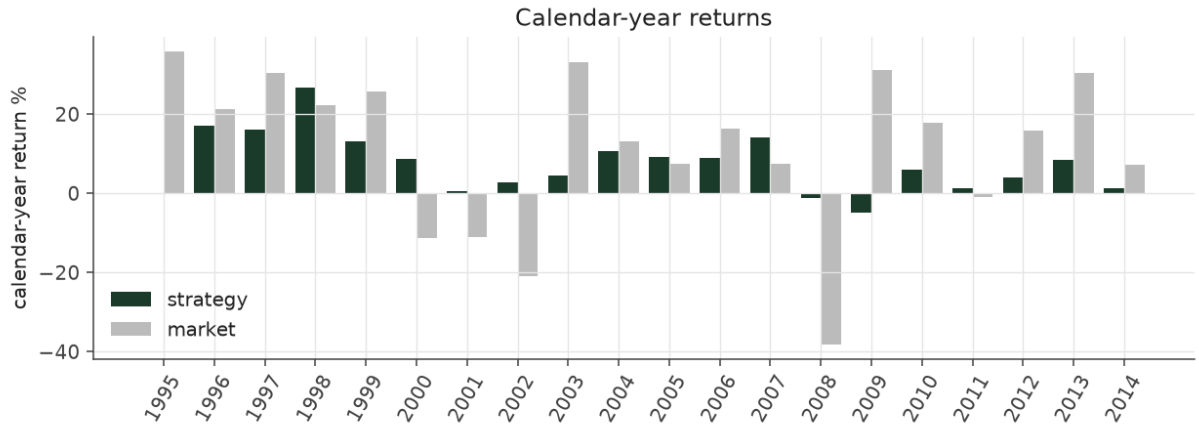
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.47



NAV (log scale)



Drawdown from running peak

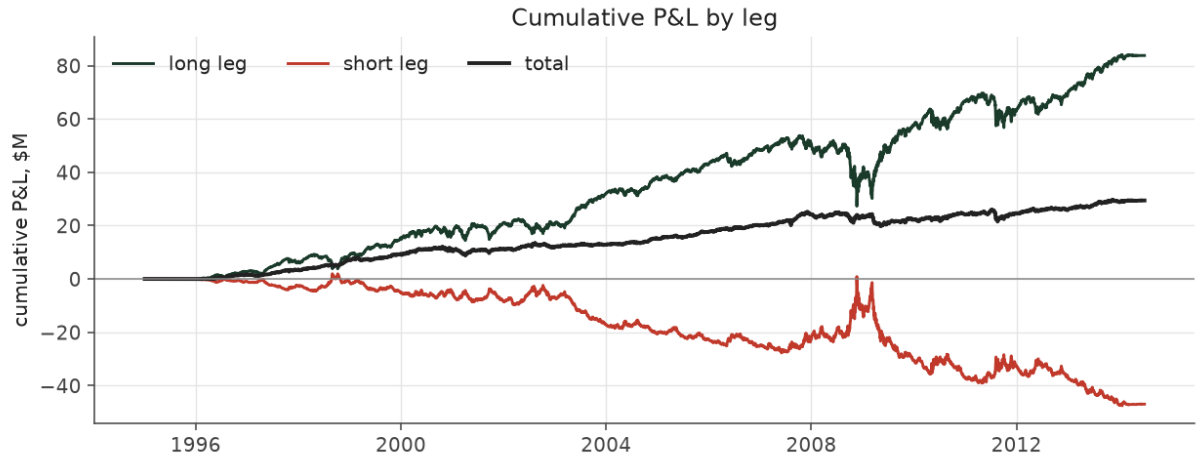


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	83.80	-46.97
P&L after costs ($M)	80.33	-50.91
of which common (average-event) component ($M)	29.35	-28.46
of which excess / PEAD component ($M)	14.08	16.52
excess minus costs ($M)	10.61	12.58
gross return per $ traded (bp)	453.74	-252.30
positions in the money (hit rate)	0.63	0.42
stand-alone Sharpe (leg only)	0.80	-0.48

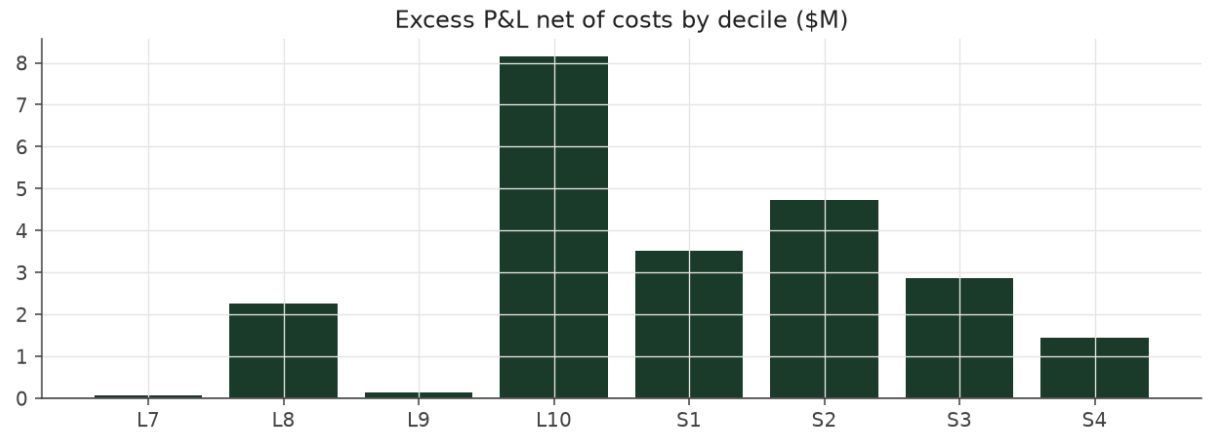
Daily P&L correlation between the two legs: **-0.97**.

• **The short leg loses money in absolute terms** (-47.0M before costs) while the long leg earns +83.8M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 29.3M / 28.5M. Stripping that common component, the surprise signal itself earned +14.1M on the long side and +16.5M on the short side.

• The two legs' daily P&L are -0.97 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.80 long / -0.48 short vs 1.11 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	151.48	6.63	0.35	0.29	0.06	0.3%	3.97
long	8	16,588	419.95	18.91	3.06	0.80	2.26	9.8%	53.88
long	9	17,747	579.02	19.40	1.20	1.07	0.13	0.6%	2.31
long	10	17,174	696.43	35.39	9.47	1.32	8.16	35.2%	117.11
short	1	17,169	687.39	-20.20	5.02	1.50	3.52	15.2%	51.23
short	2	17,281	581.80	-15.87	5.95	1.21	4.74	20.4%	81.42
short	3	15,970	427.74	-10.65	3.77	0.90	2.88	12.4%	67.22
short	4	9,171	164.64	-4.20	1.78	0.33	1.45	6.2%	87.75



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	331.20	4.24	6.76	1.26	5.50	23.7%	165.93
2	22,885	698.98	5.38	9.21	1.95	7.25	31.3%	103.78
3	25,501	855.45	6.19	7.74	1.78	5.96	25.7%	69.68
4	27,583	908.68	5.87	4.34	1.42	2.92	12.6%	32.19
5	28,947	914.14	7.73	2.56	1.00	1.56	6.7%	17.02

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	237.22	5.30	6.80	1.19	5.62	24.2%	236.80
2	23,084	737.56	4.92	8.93	2.21	6.71	28.9%	91.03
3	25,851	913.89	5.24	8.96	1.83	7.13	30.7%	78.03
4	27,665	916.54	5.74	3.57	1.28	2.28	9.8%	24.92

5	28,874	902.97	8.23	2.34	0.90	1.43	6.2%	15.87

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,393.00	-4.21	6.41	2.48	3.94	17.0%	28.25
41-60d	25,652	849.98	5.63	4.14	1.72	2.42	10.4%	28.42
61-90d	35,324	1,098.83	6.54	8.83	2.31	6.52	28.1%	59.36
>90d	12,298	366.64	21.45	11.23	0.91	10.32	44.5%	281.39

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	747.21	10.90	10.75	1.46	9.29	40.0%	124.27
2002-07	39,956	1,240.62	12.89	13.29	2.43	10.86	46.8%	87.54
2008-13	45,278	1,720.62	5.63	6.57	3.52	3.04	13.1%	17.69

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	361.52	6.68	2.57	0.76	1.81	7.8%	50.15
Chems	2,590	247.25	1.86	2.15	0.43	1.72	7.4%	69.57
Durbl	2,843	271.20	0.82	1.73	0.49	1.24	5.3%	45.72
Enrgy	6,194	306.46	2.68	5.02	0.58	4.44	19.1%	144.91
Hlth	11,696	351.27	4.78	2.91	0.74	2.17	9.4%	61.85
Manuf	12,034	328.49	2.00	2.35	0.68	1.68	7.2%	50.98
Money	17,583	324.19	2.90	3.92	0.73	3.19	13.8%	98.48
NoDur	5,081	297.45	3.35	2.32	0.56	1.75	7.6%	58.95
Other	18,377	351.79	2.42	2.83	0.78	2.05	8.8%	58.33
Shops	11,984	342.63	3.79	3.25	0.70	2.55	11.0%	74.45
Telcm	3,981	294.61	-1.67	0.89	0.54	0.35	1.5%	12.02
Utils	4,672	231.59	-0.19	0.65	0.42	0.23	1.0%	9.79

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	1,190.47	5.04	7.43	2.16	5.27	22.7%	44.28
low vol	40,091	1,262.78	13.29	12.95	2.76	10.19	43.9%	80.68
mid vol	37,903	1,199.44	10.75	9.87	2.38	7.49	32.3%	62.46

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.30%	0.00%
0.5%	1.9	7.02%	0.27%
1.0%	3.8	6.74%	0.56%
2.0%	7.6	6.12%	1.18%
5.0%	19.1	3.72%	3.58%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.